

Hedi BOUSSOFFARA

Engineering & Master's student in Telecommunications | AI/ML for Networks | Python

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PROFILE

Telecommunications engineering student (ENISO), enrolled in parallel in a Master's programme in Telecommunications, seeking a 6-month research internship in **5G network management and optimisation**. Strong background in Python, machine learning and deep learning applied to signals, time series, IoT telemetry and anomaly detection. Interested in AI/ML-driven adaptive network operations, Digital Twins and distributed learning. Fluent in English (professional level), able to analyse scientific literature, design experiments and interpret quantitative results.

EDUCATION

Engineering degree in Telecommunications

ENISO, Sousse, Tunisia

Master's degree in Telecommunications (in parallel)

ENISO, Sousse, Tunisia

TECHNICAL SKILLS

Telecommunications & networks: Wireless and mobile communications, signal processing, MQTT, WebSocket, real-time data streaming

AI / ML: Machine Learning, Deep Learning (TensorFlow/Keras, CNN, Conv1D/Conv2D), XGBoost, Scikit-learn, anomaly detection, time series, NLP, RAG

Programming & data: Python (Pandas, NumPy-based workflows), SQL, JavaScript, FastAPI, Plotly/Dash, data processing and simulation

Data & infrastructure: Kafka, Spark Streaming, PostgreSQL, MongoDB, Docker, CI/CD, Linux

Embedded & IoT: ESP32, Arduino, STM32, FPGA

RESEARCH-ORIENTED PROJECTS

5G RAN energy- and EMF-aware optimisation with a Digital Twin

Oct. 2026

- Built a 7-cell 5G simulator (3.5 GHz, 100 MHz: path loss, SINR, EARTH power model, EM-exposure proxy) and a gradient-boosting Digital Twin ($R^2 = 0.98$ on energy, 0.99 on exposure) to choose per-cell Tx power and sleep modes under a QoS constraint.
- Twin-based optimisation, validated on 120 scenarios, cut cluster energy by 25% (36% at night) and the exposure proxy by 75%, meeting the 5 Mbps QoS target in 97.5% of scenarios.
- Compared federated (FedAvg), local and centralised traffic forecasting across cells: federated MAE 1.56 UEs vs 1.57 centralised and 1.85 for persistence, without sharing raw data.
- Python, NumPy, Scikit-learn, Federated Learning, simulation, Matplotlib.*

Bearing fault detection with deep learning

Jan. 2026 - Apr. 2026

- Designed an anomaly detection system from vibration and acoustic signals; compared 1D (Conv1D) and 2D (Conv2D on GAF images) models, reaching over 90% accuracy.
- Python, TensorFlow/Keras, signal processing.*

Fleet management and AI predictive maintenance

Jan. 2026 - Apr. 2026

- Analysed vehicle and driving data to identify maintenance indicators and operational risks; built a data pipeline, a PostgreSQL database and an interactive dashboard.
- Python, XGBoost, Scikit-learn, Pandas, Dash, Plotly, PostgreSQL, SQLAlchemy.*

Real-time patient monitoring with Big Data and IoT

Sep. 2025 - Nov. 2025

- Built a real-time vital-signs collection and analysis system with automatic alerts on abnormal values.
- Kafka, Spark Streaming, MongoDB, Docker, IoT.*

ENISO student assistant with RAG

Mar. 2026 - May 2026

- Designed a RAG conversational assistant answering questions from official administrative documents, with source display and PDF download.
- Python, FastAPI, React.js, Ollama, Mistral, RAG, TF-IDF, Docker.*

PROFESSIONAL EXPERIENCE

DevSecOps Engineering Intern - Nouvelair, Monastir, Tunisia

Jul. 2026 - Sep. 2026

- Designed and developed a centralised DevSecOps platform covering 5 software lifecycle stages and integrating 7 tools.
- Automated CI/CD, security and code-quality checks (estimated 30-40% reduction in repetitive manual DevOps tasks).
- Developed an AI assistance module with Ollama to analyse pipeline failures and speed up diagnosis.

IoT and AI Developer Intern - YuccaInfo, Sousse, Tunisia

Jun. 2025 - Aug. 2025

- Developed an automated IoT supervision system (ESP32, Flask, React/Tauri, Supabase) collecting and visualising sensor data in real time.
- Integrated a prediction model to detect anomalies and anticipate trends; added AI-based decision-support functions.

CERTIFICATIONS

NVIDIA: Applications of AI for Predictive Maintenance | NVIDIA: Data Parallelism - Training Deep Learning Models on Multiple GPUs | Microsoft Azure Fundamentals (AZ-900) | AWS Certified Cloud Practitioner (CLF-C02) | LPI Linux Essentials

LEADERSHIP & ACTIVITIES

- Forum de Convergences ENISO Entreprises** (Jun. 2025 - Jun. 2026): led the Day-J committee, a team of 250+ members; coordinated logistics, partnerships and communication.
- ACM ENISO** (Sep. 2025 - May 2026): Treasurer; contributed to organising the Code Clash 2.0 programming contest.
- Aerotronix, OTC Orange Tech Club, ENISO team**: delivered introductory trainings and workshops for students.

LANGUAGES

French: fluent | **English:** professional proficiency | **German:** basic